exp 2 True map

Ground-truth-informed spatial regularisation maps for PCM U-TV
Accuracy, line retention, and PDHG computation time

PCM LION research experiment

28 October 2026

Contents

1 Executive summary **2**

2 Map construction and experimental setup **2**

2.1 Ground-truth disclosure and inference limitation 2

3 Scale calibration (calibration seeds only) **2**

4 Held-out test accuracy **3**

5 Paired changes against the calibrated U-TV baseline **4**

6 PDHG iteration count, accuracy, and runtime **4**

6.1 Line preservation 5

6.2 Fixed-depth PDHG runtime 5

6.3 Time to comparable reconstruction quality 5

6.4 Map-generation overhead 5

7 Complete four-image visual comparisons **6**

7.1 CIGS 0674 6

7.2 CIGS 0985 7

7.3 CIGS 0917 9

7.4 CIGS 0864 10

8 Interpretation and limitations **11**

9 Reproducibility and source data **12**

1 Executive summary

This experiment is an **offline counterfactual diagnostic**, not a deployable reconstruction algorithm: the actual ground-truth image is used only to compute alternative TV maps. The noisy measurements, operator, solver and initial iterate remain the same. The maps are not provably optimal; their use of ground truth must not be confused with a practical inverse-problem method. The main comparison is always against the calibrated measurement-conditioned U-TV map at $c = 8$, not only the uncalibrated $c = 1$ method. Both mean-matching and calibration-only scalar selection are used to limit confounding by global regularisation strength.

2 Map construction and experimental setup

We study $y = Ax^\star + \varepsilon$ with the same PCM Hadamard measurement simulation, sample patterns, and noise seeds as Experiment 1. The original U-TV image is produced with the learned map $\Lambda_y = \Lambda_\theta(z(y))$, where $z(y)$ is the CG image used by the trained model. The alternative CNN map is $\Lambda_{\mathrm{GT}} = \Lambda_\theta(Sx^\star)$, with S the same image intensity scaling used by the PCM forward model. Both maps use the *same frozen CNN*.

$$\widetilde{\Lambda}_{\mathrm{GT}} = \Lambda_{\mathrm{GT}} \frac{\mathrm{mean}(\Lambda_y)}{\mathrm{mean}(\Lambda_{\mathrm{GT}})},$$
$$\Lambda_{\mathrm{edge}} = \mathrm{mean}(\Lambda_y) \frac{w(\nabla S x^\star)}{\mathrm{mean}(w(\nabla S x^\star))},$$
$$w(g) = \epsilon + (1-\epsilon)\exp[-(g/\kappa)^2].$$

Here $\epsilon = 0.05$ and κ is the 90th percentile of strictly positive GT gradient magnitudes. The edge map is designed to penalise known sharp transitions less strongly; it is a heuristic, *not* an oracle-minimising parameter map. Mean-matching makes the average unscaled TV penalty comparable to the measurement-conditioned baseline.

$$x_{m,c,K} = \mathrm{PDHG}_K(y,\ c\Lambda_m)\,,$$

with identical y, A, zero initialisation and model step sizes for all compared maps. All primary reconstructions use 256 steps. Map-scale candidates are selected only from the two calibration noise seeds, and the results below are for three held-out noise seeds per image.

2.1 Ground-truth disclosure and inference limitation

Using $x^\star$ to design a regulariser can produce more informative spatial weights but is unavailable for real measurements. Any apparent reduction in pre-processing time by bypassing CG is *not* a realistic inference-time improvement. GPU timing on a 4070 shared with ongoing model training is exploratory until repeated on an otherwise idle device.

3 Scale calibration (calibration seeds only)

Map type	Selected scalar c	Calibration mean PSNR (dB)
gt_cnn_meanmatched	8.000	32.139
gt_edge_meanmatched	0.500	31.323

This selection uses ground-truth PSNR for the two calibration-noise seeds of each of the same four CIGS images. The three held-out test-noise seeds are excluded from selecting scale; this is not a test

on previously unseen image structures. **Boundary warning:** the GT-edge map selects $c = 0.5$, the *smallest* prespecified candidate in $\{0.5, 1, 2, 4, 8, 16, 32\}$. Its optimum may lie below the tested range. Its observed PSNR deficit therefore does not rule out better edge-map performance with a wider scale search.

4 Held-out test accuracy

Method	PSNR	SSIM	HaarPSI	MSE $\times 10^3$
measured_c1	28.236	0.812	0.780	1.503
measured_c8	32.240	0.949	0.864	0.599
gt_cnn_raw_c1	28.453	0.817	0.768	1.430
gt_cnn_raw_c8	32.217	0.951	0.858	0.602
gt_cnn_matched_calibrated	32.095	0.950	0.866	0.619
gt_edge_matched_calibrated	31.367	0.911	0.830	0.732

Means are balanced across the four images, each evaluated at three noise seeds. With only four distinct images, do not interpret noise-seed variations as twelve independent image samples.

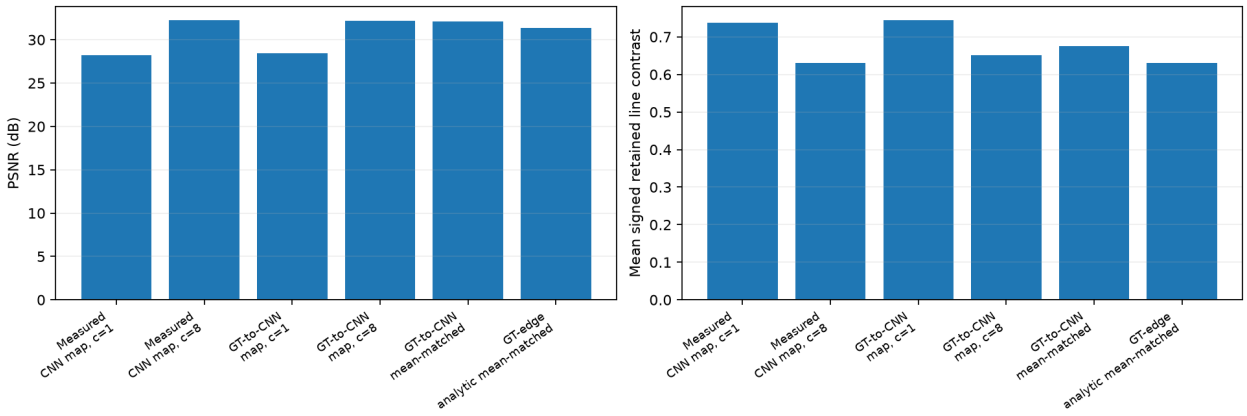


Figure 1: Global accuracy and local annotated-line contrast. A gain in PSNR can coexist with a loss in a true thin line.

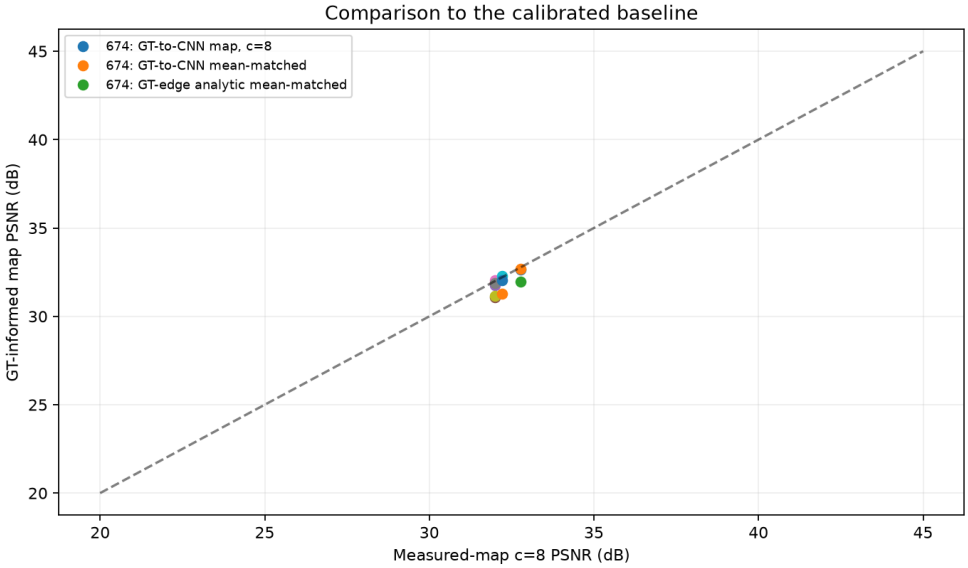


Figure 2: Comparison of ground-truth-assisted maps against the calibrated $c = 8$ measurement-conditioned map. Points above the diagonal favour the alternative map.

5 Paired changes against the calibrated U-TV baseline

Method	Mean PSNR difference (dB)
measured_c1	-4.0035
measured_c8	+0.0000
gt_cnn_raw_c1	-3.7865
gt_cnn_raw_c8	-0.0226
gt_cnn_matched_calibrated	-0.1450
gt_edge_matched_calibrated	-0.8728

The signed line contrast is evaluated separately. CIGS image 0917 contains a conductor track, not a confirmed crack; none of the four line annotations is a validated material-defect segmentation.

6 PDHG iteration count, accuracy, and runtime

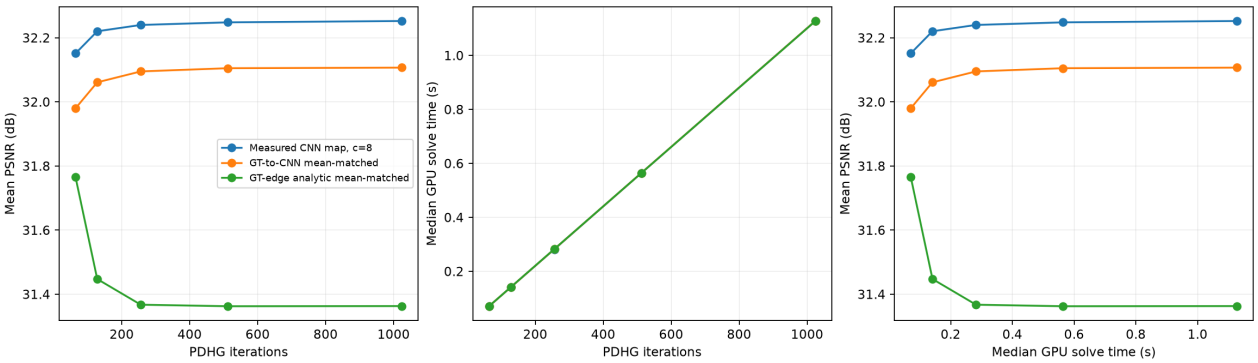


Figure 3: Measured PDHG iterations, median solve time and image accuracy. Timings exclude GT acquisition and are not a deployable end-to-end speed comparison.

Time-to-quality is measured by the earliest step count reaching the measured-map $c = 8$ PSNR (256 steps) minus 0.1 dB. A second criterion measures distance to each map's own 1024-step approximate solution. Neither is a rigorous primal-dual gap certificate. Methods may fail to reach a common target; these failures are reported rather than assigned a fictional runtime. All whole-image and ROI comparisons use a common intensity range $[0, 1.2]$. The maps in each per-image figure use a shared range across displayed variants; mean-matched maps suppress the effect of arbitrary CNN output scale. These figures show *one predeclared test noise realisation per image*; the quantitative tables cover all three.

6.1 Line preservation

Method	Retained contrast	Line recall	False positive rate
measured_c1	0.738	0.845	0.151
measured_c8	0.631	0.750	0.045
gt_cnn_raw_c1	0.744	0.853	0.142
gt_cnn_raw_c8	0.652	0.750	0.034
gt_cnn_matched_calibrated	0.676	0.750	0.042
gt_edge_matched_calibrated	0.631	0.753	0.061

These are fixed-threshold fine-line diagnostics, not verified crack-segmentation metrics. Image 0917 contains a conductor track.

6.2 Fixed-depth PDHG runtime

Method	128 steps (s)	256 steps (s)	512 steps (s)
measured_c8	0.1410	0.2816	0.5639
gt_cnn_matched_calibrated	0.1410	0.2819	0.5642
gt_edge_matched_calibrated	0.1412	0.2823	0.5640

Times are measured from the actual GPU execution and exclude GT acquisition and CG construction. Concurrent 4070 training may affect timings.

6.3 Time to comparable reconstruction quality

Method	Successful / 12	Median steps	Median time (s)
measured_c8	12/12	96	0.1056
gt_cnn_matched_calibrated	3/12	128	0.1408
gt_edge_matched_calibrated	0/12	–	–

The reference is the measured-map c=8 at 256 iterations, allowing 0.1 dB lower PSNR. Inaccessible quality targets are failures, not zero-cost successes.

6.4 Map-generation overhead

Map creation	Median seconds
CNN(CG(y))	0.000499
CNN($Sx^\star$)	0.000467
GT edge weighting	0.001455

These costs assume the input image is already available. GT itself is inaccessible in real PCM reconstruction and cannot be counted as a speed gain.

7 Complete four-image visual comparisons

7.1 CIGS 0674

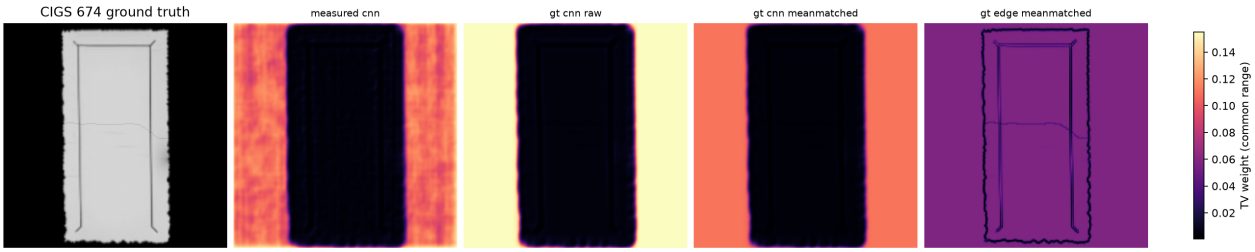


Figure 4: GT and spatial regularisation maps for CIGS 0674. The GT-informed CNN prediction is not an optimal or deployable map.

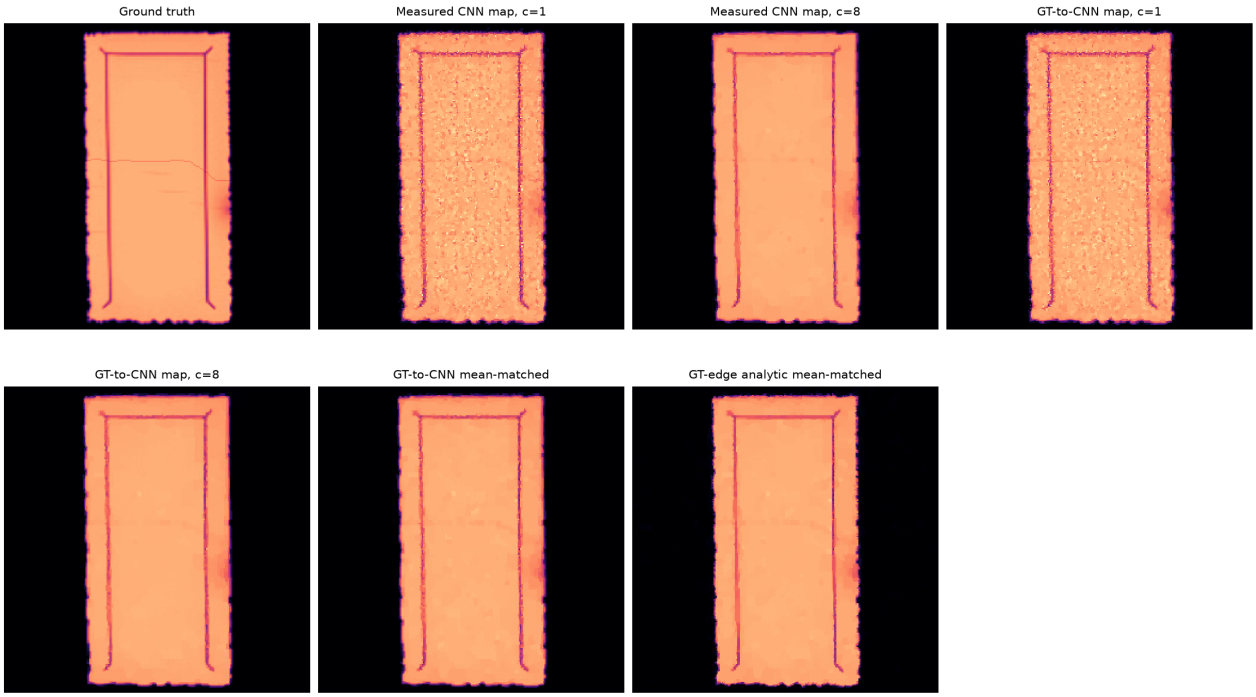


Figure 5: Whole-image reconstruction comparison for CIGS 0674 under one shared acquisition.

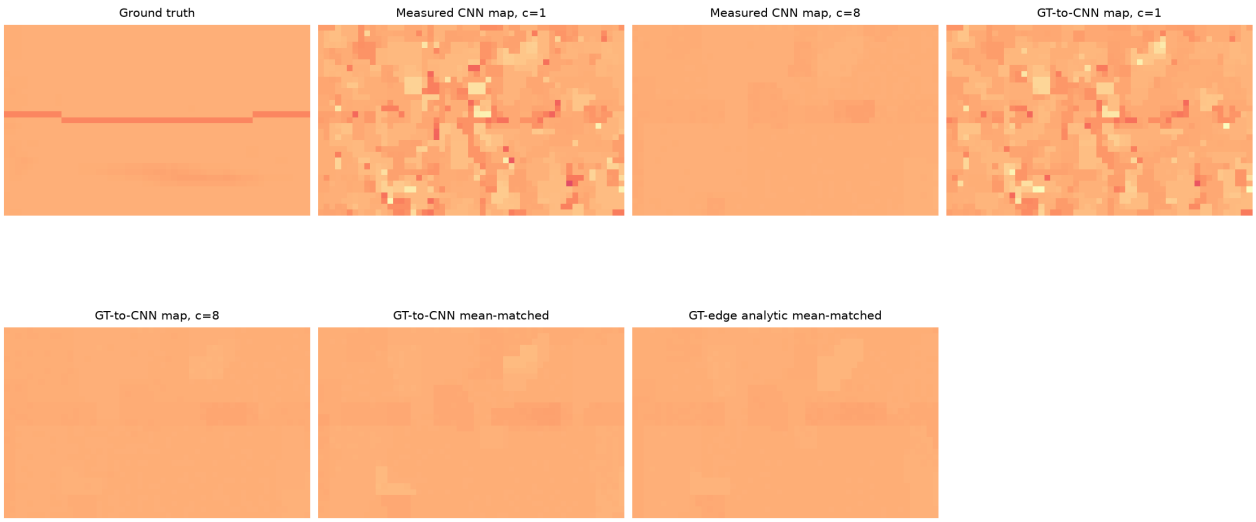


Figure 6: Common-scale close-ups of the audited fine-line region, CIGS 0674. Distinguish the conductor track in image 0917 from crack-like lines.

7.2 CIGS 0985

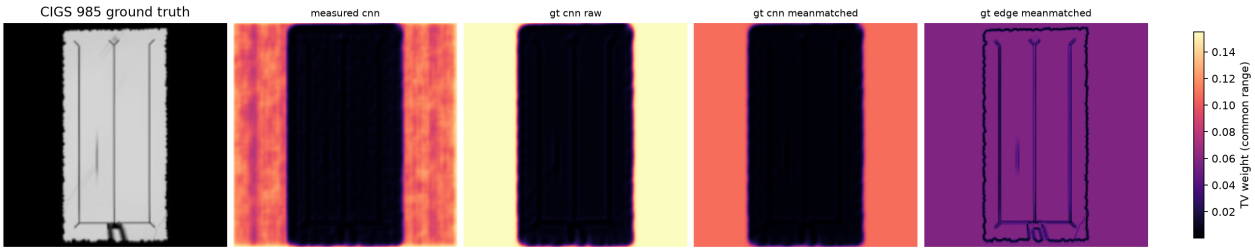


Figure 7: GT and spatial regularisation maps for CIGS 0985. The GT-informed CNN prediction is not an optimal or deployable map.

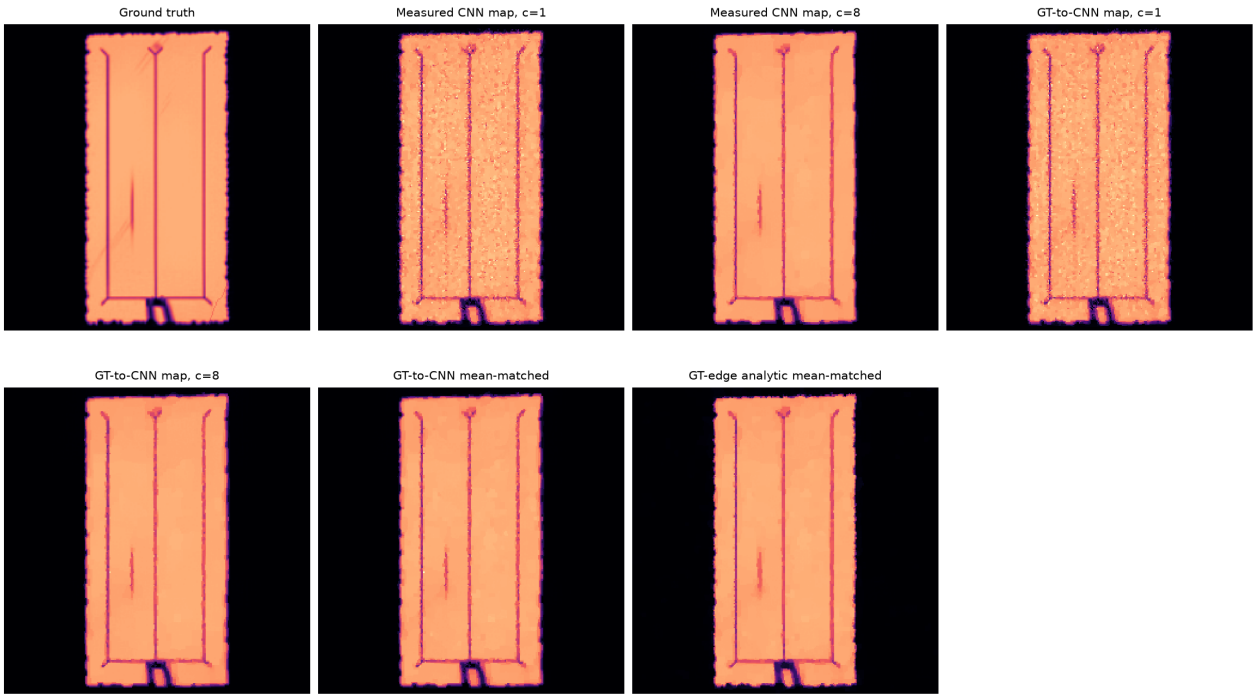


Figure 8: Whole-image reconstruction comparison for CIGS 0985 under one shared acquisition.

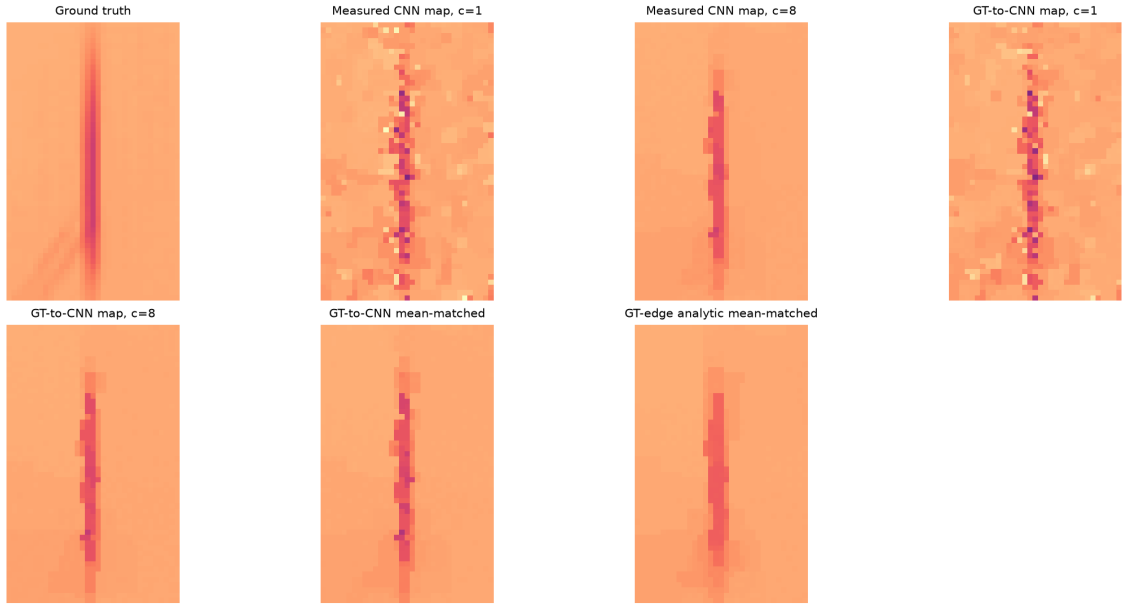


Figure 9: Common-scale close-ups of the audited fine-line region, CIGS 0985. Distinguish the conductor track in image 0917 from crack-like lines.

7.3 CIGS 0917

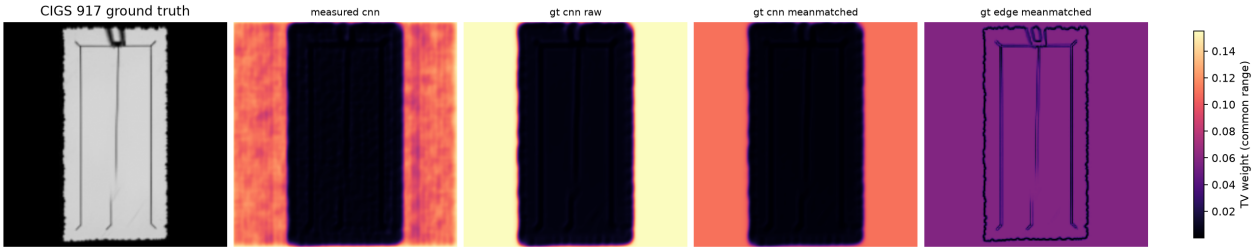


Figure 10: GT and spatial regularisation maps for CIGS 0917. The GT-informed CNN prediction is not an optimal or deployable map.

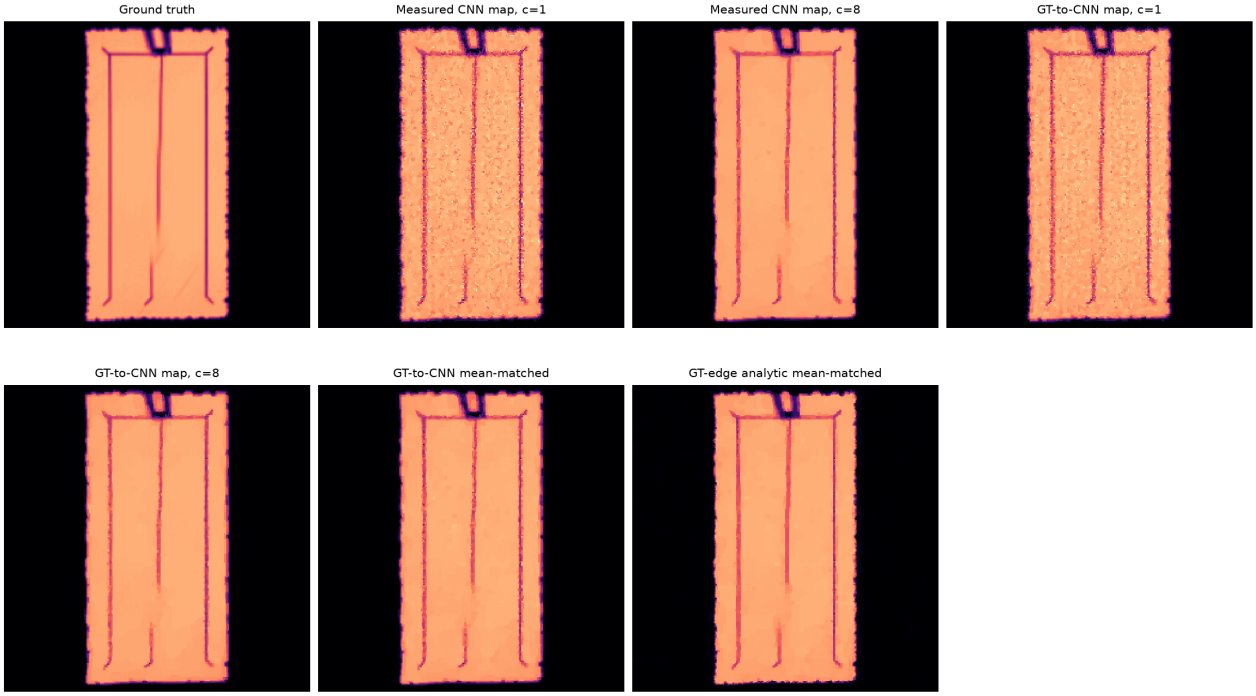


Figure 11: Whole-image reconstruction comparison for CIGS 0917 under one shared acquisition.

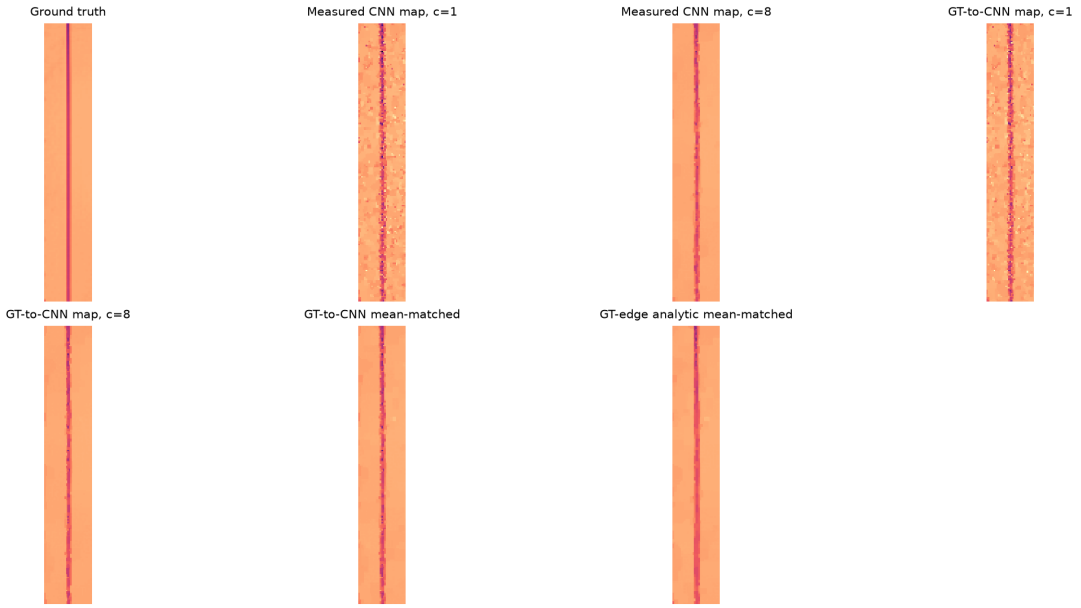


Figure 12: Common-scale close-ups of the audited fine-line region, CIGS 0917. Distinguish the conductor track in image 0917 from crack-like lines.

7.4 CIGS 0864

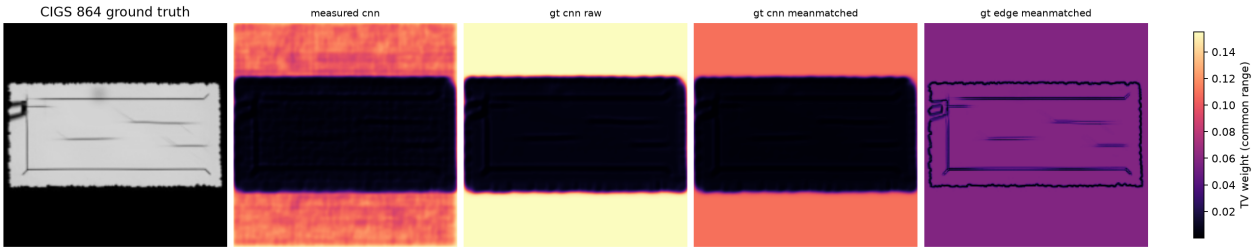


Figure 13: GT and spatial regularisation maps for CIGS 0864. The GT-informed CNN prediction is not an optimal or deployable map.

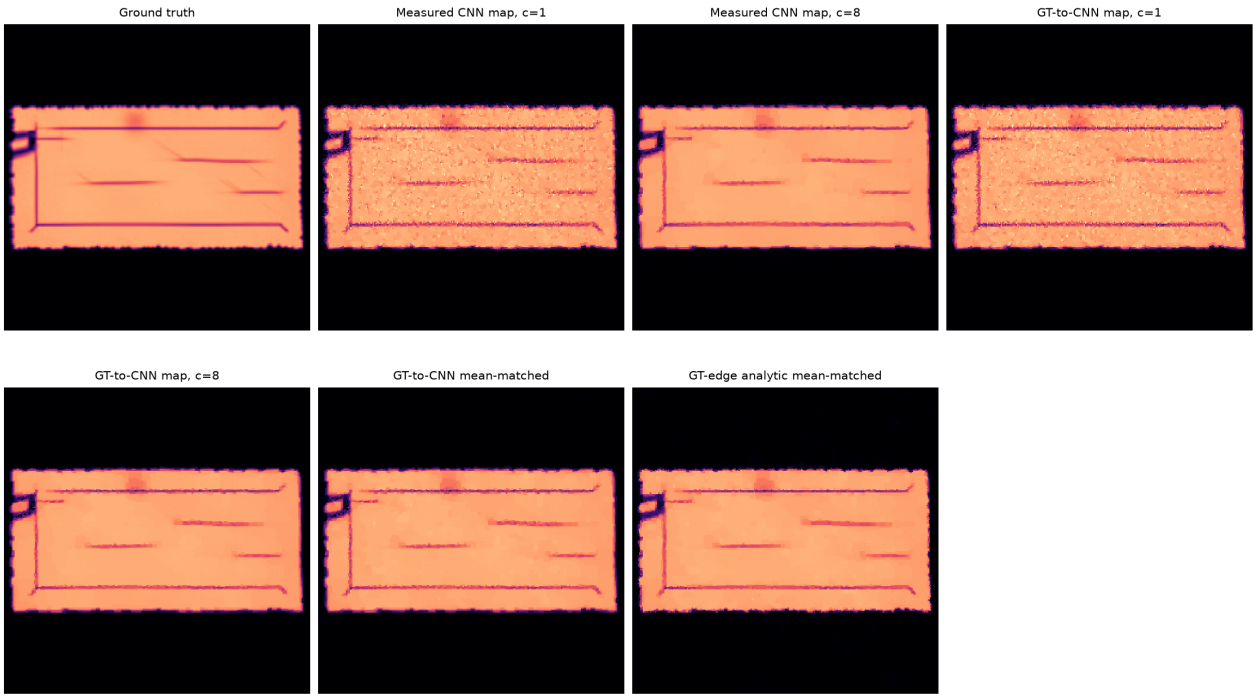


Figure 14: Whole-image reconstruction comparison for CIGS 0864 under one shared acquisition.

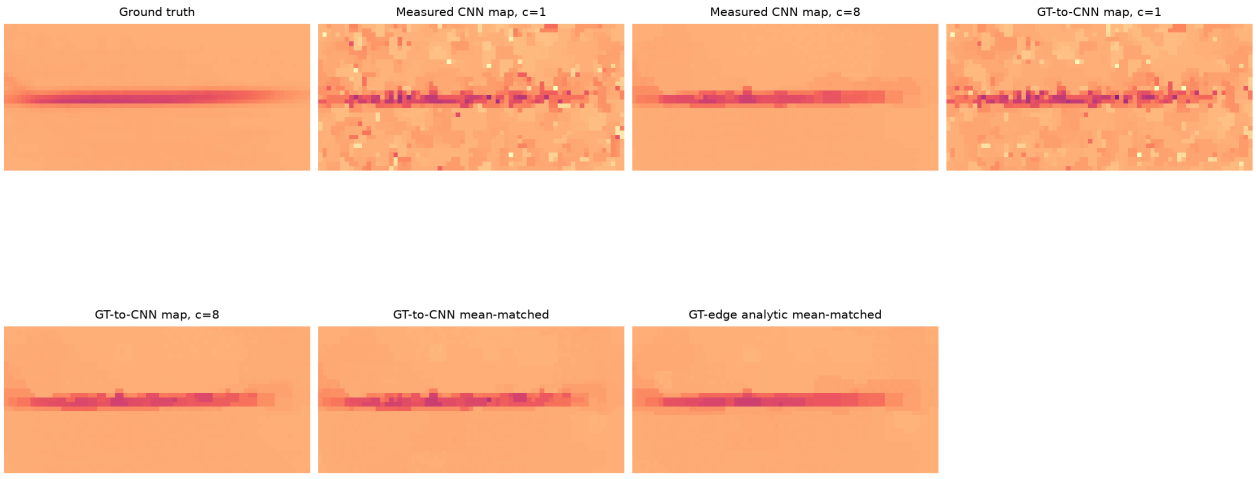


Figure 15: Common-scale close-ups of the audited fine-line region, CIGS 0864. Distinguish the conductor track in image 0917 from crack-like lines.

8 Interpretation and limitations

1. The GT-informed map need not minimise expected reconstruction error, because the original U-Net was never trained to map clean ground truth to optimal regularisation weights. Passing GT may also be an out-of-distribution input relative to its CG training images.

2. The GT-edge map is a hand-crafted spatial prior; matching global map means is helpful but does not imply equal effective TV regularisation for each image. Its selected scalar is the lower

boundary of the tested search grid; sub-0.5 scalars are an untested sensitivity analysis.

3. Compare measured $c = 8$ with GT-informed variants at the same PDHG depth. At a fixed number of iterations, a changed map need not make PDHG itself faster.

4. A comparison at the same target PSNR can show practical iteration savings, but it is not a mathematically rigorous convergence-rate comparison because each map defines a different objective.

5. Input GT is supplied for free only as an upper-bound diagnostic. It must not be described as a deployable PCM measurement-to-reconstruction algorithm or measured as a real reduction in end-to-end inference latency.

6. These four images were already examined in Experiment 1. The result is exploratory; scale-calibration seeds and test-noise seeds are disjoint, but no completely untouched image population is claimed.

9 Reproducibility and source data

All experiment code, source protocol, map arrays, reconstructions and metrics were generated with a frozen checkpoint and the exact Experiment 1 acquisition seeds. The archive includes `protocol.json`, `metrics.csv`, `selection.json`, and report figures. No new network parameters were trained.